

$$\begin{aligned} & \frac{1}{16\pi^2} \rightarrow \left(-\frac{1}{6} g_2^4 \text{LF}_{3,0} [m_2] - \frac{1}{4} g_2^4 \text{LF}_{4,-1} [m_2] + \frac{4}{15} g_2^4 \text{LF}_{5,-2} [m_2] - \frac{1}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_{\mathbf{d}}^{\mathbf{p}}] + \right. \\ & \frac{1}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_{\mathbf{d}}^{\mathbf{p}}] + \frac{1}{2} g_1^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,0} [m_{\mathbf{d}}^{\mathbf{r}}] - \frac{1}{2} g_1^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{4,-1} [m_{\mathbf{d}}^{\mathbf{r}}] - \\ & \frac{1}{8} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_{\mathbf{e}}^{\mathbf{p}}] + \frac{1}{8} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_{\mathbf{e}}^{\mathbf{p}}] + \frac{1}{2} g_1^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,0} [m_{\mathbf{e}}^{\mathbf{r}}] - \\ & \frac{1}{2} g_1^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{4,-1} [m_{\mathbf{e}}^{\mathbf{r}}] + \frac{1}{48} (-12 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} (g_1^2 + 2 g_2^2) + \sum_{\mathbf{p}} (-3 g_1^4 + 2 g_2^4)) \text{LF}_{3,0} [m_{\mathbf{l}}^{\mathbf{p}}] + \\ & \frac{1}{32} (8 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} (g_1^2 + 2 g_2^2) + \sum_{\mathbf{p}} (2 g_1^4 + g_2^4)) \text{LF}_{4,-1} [m_{\mathbf{l}}^{\mathbf{p}}] - \\ & \frac{1}{15} \sum_{\mathbf{p}} g_2^4 \text{LF}_{5,-2} [m_{\mathbf{l}}^{\mathbf{p}}] + \left(\frac{1}{4} \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (g_1^2 - 6 g_2^2) - \frac{1}{48} \sum_{\mathbf{p}} (g_1^4 - 6 g_2^4) \right) \text{LF}_{3,0} [m_{\mathbf{q}}^{\mathbf{p}}] + \\ & \frac{1}{96} (-24 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (g_1^2 - 6 g_2^2) + \sum_{\mathbf{p}} (2 g_1^4 + 9 g_2^4)) \text{LF}_{4,-1} [m_{\mathbf{q}}^{\mathbf{p}}] - \frac{1}{5} \sum_{\mathbf{p}} g_2^4 \text{LF}_{5,-2} [m_{\mathbf{q}}^{\mathbf{p}}] - \\ & \frac{1}{6} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_{\mathbf{u}}^{\mathbf{p}}] + \frac{1}{6} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_{\mathbf{u}}^{\mathbf{p}}] - \frac{1}{12} g_2^4 \text{LF}_{3,0} [\tilde{\mu}] - \frac{1}{8} g_2^4 \text{LF}_{4,-1} [\tilde{\mu}] + \\ & \frac{2}{15} g_2^4 \text{LF}_{5,-2} [\tilde{\mu}] - \frac{7}{32} g_1^4 \text{LF}_{2,2,-1} [m_1, \tilde{\mu}] - \frac{1}{16} g_1^4 m_1^2 \text{LF}_{2,2,0} [m_1, \tilde{\mu}] + \frac{1}{2} g_1^4 \text{LF}_{3,2,-2} [m_1, \tilde{\mu}] + \\ & \frac{3}{8} g_1^4 m_1^2 \text{LF}_{3,2,-1} [m_1, \tilde{\mu}] - \frac{1}{4} g_1^4 \text{LF}_{3,3,-3} [m_1, \tilde{\mu}] - \frac{1}{4} g_1^4 m_1^2 \text{LF}_{3,3,-2} [m_1, \tilde{\mu}] - \\ & \frac{1}{4} g_1^4 \text{LF}_{4,2,-3} [m_1, \tilde{\mu}] - \frac{1}{4} g_1^4 m_1^2 \text{LF}_{4,2,-2} [m_1, \tilde{\mu}] + g_2^4 \text{LF}_{2,1,0} [m_2, \tilde{\mu}] + \frac{33}{32} g_2^4 \text{LF}_{2,2,-1} [m_2, \tilde{\mu}] - \\ & \frac{1}{16} g_2^4 m_2^2 \text{LF}_{2,2,0} [m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 \text{LF}_{3,1,-1} [m_2, \tilde{\mu}] - g_2^4 \text{LF}_{3,2,-2} [m_2, \tilde{\mu}] + \\ & \frac{21}{8} g_2^4 m_2^2 \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] + \frac{1}{4} g_2^4 \text{LF}_{3,3,-3} [m_2, \tilde{\mu}] - \frac{7}{4} g_2^4 m_2^2 \text{LF}_{3,3,-2} [m_2, \tilde{\mu}] + \\ & \frac{1}{4} g_2^4 \text{LF}_{4,2,-3} [m_2, \tilde{\mu}] - \frac{7}{4} g_2^4 m_2^2 \text{LF}_{4,2,-2} [m_2, \tilde{\mu}] - \frac{3}{2} \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{2,1,0} [m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{d}}^{\mathbf{t}}] + \\ & \frac{3}{2} \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{3,1,-1} [m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{d}}^{\mathbf{t}}] + \frac{1}{2} g_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,1,0} [m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] - \\ & \frac{1}{4} g_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] - \frac{1}{2} g_1^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{4,1,-1} [m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] - \\ & \frac{1}{2} \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{2,1,0} [m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{e}}^{\mathbf{t}}] + \frac{1}{2} \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{3,1,-1} [m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{e}}^{\mathbf{t}}] + \\ & \frac{1}{2} g_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,1,0} [m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{l}}^{\mathbf{p}}] - \frac{1}{4} g_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{l}}^{\mathbf{p}}] - \\ & \frac{1}{2} g_1^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{4,1,-1} [m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{l}}^{\mathbf{p}}] - \frac{1}{8} \overline{a_e^{\text{pr}}} a_e^{\text{pr}} (g_1^2 + 5 g_2^2) \text{LF}_{4,1,-1} [m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] + \\ & \frac{1}{2} g_2^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{5,1,-2} [m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] + \frac{1}{4} \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{2,1,0} [m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{l}}^{\mathbf{s}}] - \\ & \frac{1}{4} \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{3,1,-1} [m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{l}}^{\mathbf{s}}] + \frac{1}{8} \overline{a_d^{\text{pr}}} a_d^{\text{pr}} (g_1^2 - 15 g_2^2) \text{LF}_{4,1,-1} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{d}}^{\mathbf{r}}] + \\ & \frac{3}{2} g_2^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{5,1,-2} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{d}}^{\mathbf{r}}] + \frac{3}{4} \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{2,1,0} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{q}}^{\mathbf{s}}] - \\ & \frac{3}{4} \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \text{LF}_{3,1,-1} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{q}}^{\mathbf{s}}] - \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{2,2,0} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{u}}^{\mathbf{r}}] + \\ & \frac{1}{4} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + 6 g_2^2) \text{LF}_{3,1,0} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{u}}^{\mathbf{r}}] - \frac{1}{8} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + 3 g_2^2) \text{LF}_{3,2,-1} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{u}}^{\mathbf{r}}] - \\ & \frac{1}{4} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + 6 g_2^2) \text{LF}_{4,1,-1} [m_{\mathbf{q}}^{\mathbf{p}}, m_{\mathbf{u}}^{\mathbf{r}}] + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,1,0} [m_{\mathbf{u}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] + \\ & \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_{\mathbf{u}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] - \frac{1}{4} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (2 g_1^2 + 9 g_2^2) \text{LF}_{4,1,-1} [m_{\mathbf{u}}^{\mathbf{r}}, m_{$$